Dexter Williams

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Education

University of Wisconsin-Madison *Madison, Wisconsin*

PhD in Information Science *Aug. 2025 - Present*

- Advisor: Professor Jodi Schneider.
- Research Interests: Argument Mining, Computational Argumentation, AI and Law.
- 4.0 GPA

University of Illinois Urbana-Champaign *Champaign, Illinois*

PhD in Informatics *Aug. 2024 - July 2025*

- Advisor: Professor Jodi Schneider.
- 4.0 GPA

University of St Andrews *St Andrews, United Kingdom*

BSc in Computer Science and Philosophy (Joint Honours) *Sep. 2018 - May 2022*

- First Class (3.7-3.9 GPA).

Skills

Programming Languages (Strong) Java, Python, Ada, C, Lua, JavaScript/TypeScript.

Programming Languages (Other) Prolog, SQL, Haskell, R.

General Software engineering, Agile, NLP.

Experience

University of Wisconsin-Madison *Madison, Wisconsin, United States*

Project Assistant (Professor Jodi Schneider) *Aug. 2025 - Present*

- Wrote an appendix for a book on argument mining, that documents existing corpora, and developed an website equivalent of the appendix that can be contributed to by the argument mining community.
- Acquired copyright permissions for the 100+ figures used in the book, following up with authors with requests for republication rights.
- Supervised/mentored an undergraduate who provided support for the corpora appendix.

Boise State University *Boise, Idaho, United States*

Research Intern *Jun. 2025 - Jul. 2025*

- Joined the Retico team at Boise State University for 1 month to aid in the research and development of state-of-the-art incremental processing systems.
- Developed two new modules for Retico to handle speech diarization and automatic speech recognition.
- Implemented an end-to-end incremental processing project that performs real-time argument mining on a debate to produce an abstract argumentation framework.

University of Illinois Urbana-Champaign *Champaign, Illinois, United States*

Graduate Research Assistant (Professor Jodi Schneider) *May 2025 - Aug. 2025*

- Provided feedback and review for a new edition of a book on argumentation mining.
- Read and categorized 100+ papers on argument mining to assist in the writing of a new chapter.
- Explored methods for classifying the multitude of existing argument mining corpora.

University of Illinois Urbana-Champaign *Champaign, Illinois, United States*

Graduate Research Assistant (Professor Bertram Ludäscher) *May 2025 - Aug. 2025*

- Joined the explainable AI through computational argumentation (XAI-CA) project, which explores how argumentation can aid in explainability.
- Created a program in Alloy to generate minimal abstract argumentation frameworks satisfying certain properties.
- Applied relation-based argument mining to US-legal documents, to automatically construct structured argumentation frameworks.

University of Illinois Urbana-Champaign *Champaign, Illinois, United States*

Teaching Assistant *Aug. 2024 - May. 2025*

- Assisted in teaching 80+ students introductory formal logic, discrete mathematics, and statistics.
- Ran weekly office hours.
- Graded biweekly assignments.

Capgemini Engineering *Bath, England, United Kingdom*

Associate Software Engineer / Consultant *Jun. 2023 - Jul. 2024*

- Joined the development team for a large rail project.
- Assisted in writing and testing the Alloy specification of the project.
- Produced Ada SPARK implementation based on the formal specification.
- Tested the project using a suite of tools developed under the HICLASS Research Programme.
- Provided continuous feedback on the test tools to improve their usability.

Capgemini Engineering *Bath, England, United Kingdom*

Junior Software Engineer / Consultant *Sep. 2022 - Jun. 2023*

- Designed a prototype for an engineering design tool.
- Developed the tool, creating an application with a React.js + Redux.js + Plotly.js frontend, and a Python + Flask backend.
- Used Redux Thunk to manage the complicated asynchronous state of the prototype.
- Communicated weekly with the clients of the project, and biweekly with end-users, to gather feedback.
- Organized workload into tickets based on gathered feedback.
- Presented a final demo of the prototype which received positive feedback from the end-user community.

NHS Digital *Leeds, United Kingdom (remote)*

Data Science and Innovation Summer Intern *Jul. 2021 - Sep. 2021*

- Joined the RAP (reproducible analytical pipelines) team at NHS Digital for ten weeks.
- Produced four guides that teach good software engineering practices in order to help teams update their projects to follow RAP guidelines.
- Investigated existing PySpark libraries to find a simple and reproducible way for teams to implement basic metadata input validation in their projects.
- Engaged in daily stand-up meetings, sprint planning, and retrospective meetings. Met and spoke with colleagues outside of work in order to become a good team member.
- Received a recommendation at the end of my internship for meeting/exceeding all my objectives.

University of St Andrews *St Andrews, United Kingdom*

Undergraduate Research Assistant (Professor Ian Gent) *Oct. 2020 - Feb. 2021*

- Assisted in the progress of a research project which aims to provide human explanations of solutions to Sudoku problems.
- Responsible for investigating variants of Sudoku and testing the effectiveness of the project in solving them.
- Communicated concerns and discussed the project with my supervisor as well as three other members of research staff.

University of St Andrews *St Andrews, United Kingdom*

Undergraduate Research Assistant (Dr. Kasim Terzic) *Oct. 2019 - May. 2020*

- Participated in the development of a program to be used by the Marine Biology department. The program uses machine learning to identify hundreds of seals in large aerial photographs.
- Liaised with my project supervisor every week to seek assistance in understanding and developing the project.
- Updated the output of the program to give it practical use by the given deadline.

Publications

Williams, D. (Forthcoming). That Feels Convincing: A Taxonomy Of Pathotic Argumentation Quality, Proceedings of the 5th European Conference on Argumentation *Forthcoming*

Williams, D. (Forthcoming). Computational Assessment of Pathotic Argumentation Quality, Online Handbook of Argumentation for AI, Volume 5 *Forthcoming*

Williams, D. (Forthcoming). Decorum in the Forum: Emotional Interactions in U.S. Presidential Press Conferences, Arguing Democracy *Forthcoming*

Zheng, H., **Williams, D.**, & Ludäscher, B. (2025). Using LLMs to Model Arguments in U.S. Supreme Court Briefs: Preliminary Report, Proceedings of the International Workshop on Translating Natural Legal Language Into Formal Representations *2025*